AegisZero: Complete Enterprise Project Dossier

Autonomous Zero-Trust Identity Governance & SaaS FinOps Swarm | 12-Minute Keynote Blueprint

Hackathon: Workato Hack: Agentic India 2026	**Target Presentation Duration:** 12 Minutes Full Keynote
Track: Track 1: Enterprise Agentic Workflows	**Team Lead & Presenter:** Saicharan Adapa & Team
Live Cockpit: https://aegiszero-cockpit.vercel.app	**Workato Recipe ID:** 82962492 (Active Cloud Backend)
Embedded Database: Workato Data Tables (Enterprise_Identities)	**Core Frameworks:** SOC2 Type II CC6.1, ISO 27001, AST CI/CD Static Proof

1. Enterprise Challenge & Industry Problem Statement

The Fundamental Dilemma: In enterprise organizations, employees frequently transition between squads—such as moving from the Payment Gateway squad to Internal Tooling. While new tools are provisioned immediately, historical credentials (AWS Root keys, Figma Enterprise seats, Datadog Admin tokens) are never revoked. IT departments deliberately avoid manual deprovisioning due to an acute fear of breaking background automated processes, CI/CD pipelines, or cron scripts, triggering costly production downtime.

The Anatomy of 'Toxic Admin Keys':

- **Compliance Penalties:** Fails SOC2 CC6.1 and ISO 27001 Annex A.9.2.6 segregation of duties, risking enterprise certifications and vendor trust.
- **FinOps License Waste:** Burns an average of $55,500.00 / year in zombie license fees per 100 monitored employees across inactive SaaS tiers.
- **Root Breach Threat:** 81% of cloud enterprise security breaches stem from compromised dormant standing administrative credentials.
- **Human Review Burnout:** SecOps teams spend 120+ hours each quarter reviewing manual spreadsheets and pinging 50+ managers across Slack.

2. Deep-Dive: The 8 Specialized Swarm Agents & Security Checks

AegisZero solves this challenge through an 8-agent autonomous consensus mesh. Rather than relying on blunt scripts, every deprovisioning decision passes through mathematical proofs and automated security guardrails:

• Agent 1: HR Context & Role Drift Telemetry Agent

Mechanism: Ingests HRIS records (Workday, BambooHR) and maps squad changes. Detects when an employee has switched teams and calculates tenure drift days (e.g. David Chen: 75 days post-transfer). Identifies orphaned approval chains and previous manager sign-offs.

Autonomous Output: Flag: ROLE_DRIFT_CRITICAL (75 days since squad transition; old squad approvals obsolete).

• Agent 2: Work Entropy & Behavioral Activity Audit Agent

Mechanism: Performs read vs. write behavioral telemetry. Audits GitHub commits, PRs, database query logs, and Figma canvas modifications. Cross-references logins against real production edits to identify 'passive lurkers' who login via SSO to appear active but have 0 contributions in 60+ days.

Autonomous Output: Score: 94% Dormancy (0 commits, 0 PRs, 0 Figma edits in 60 days; $1,476/mo wasted).

• Agent 3: AST & CI/CD Pipeline Blast-Radius Emulator Agent

Mechanism: The critical mathematical safety breakthrough. Parses Abstract Syntax Trees (AST) of 42 Jenkins, GitHub Actions, Dockerfiles, and cron scripts. Inspects environment secrets and IAM token dependencies to prove whether revoking standing keys will break any build pipeline.

Autonomous Output: Safety Proof: **Blast Radius = 0.0% (Zero Dependencies Found; 100% Outage-Safe)**.

• Agent 4: FinOps Arbitrage & Micro-Tiering Downgrade Agent

Mechanism: Synthesizes graceful tier downgrades instead of blunt deprovisioning. Downgrades AWS Root ($1200/mo) to 1-hour STS temporary roles ($0), Figma Pro ($75/mo) to Free Collaborator ($0), and Datadog Admin ($180/mo) to Standard Viewer ($0).

Autonomous Output: Reclaimed Value: **+$17,700/yr per user; +$55,500.00/yr across enterprise directory**.

• Agent 5: Zero-Trust Security & SOC2 Compliance Auditor Agent

Mechanism: Audits entitlements against SOC2 Type II CC6.1 and ISO 27001 Annex A.9.2.6 controls. Identifies toxic segregation of duties violations and flags standing root privilege. Dynamically re-evaluates compliance posture once keys are neutralized.

Autonomous Output: Compliance Transformation: **SOC2 posture surges from 18% (Critical) to 98% (Audit-Ready)**.

• Agent 6: Ephemeral JIT & SCIM Payload Synthesizer Agent

Mechanism: Generates deterministic API payloads to execute least-privilege neutralization. Produces AWS CLI `detach-user-policy` commands, creates JIT assume-role IAM policies, and synthesizes Figma/Datadog SCIM micro-tiering API patches.

Autonomous Output: Orchestration Payloads: Detaches AdministratorAccess; enables 1-hour temporary STS roles.

• Agent 7: Self-Healing Employee Slack Bot Comms Agent

Mechanism: Powers frictionless employee communication via the Workato Slack Connector. Delivers formatted cards to the employee and security channels with interactive 1-click 4-hour JIT elevation controls to prevent helpdesk ticketing delays.

Autonomous Output: ChatOps Impact: Zero friction for employees; instant emergency self-service access.

• Agent 8: Immutable SOC2 Audit Ledger Scribe Agent

Mechanism: Generates cryptographically signed audit attestations. Calculates a SHA-256 ledger hash binding the employee ID, AST proof, Workato execution receipt, and compliance standard, providing irrefutable evidence for KPMG/PwC auditors.

Autonomous Output: Attestation Hash: Permanent SHA-256 cryptographic proof exportable as JSON certificate.

3. Workato Stack Implementation & Pipeline Architecture

AegisZero is natively integrated with the Workato Enterprise Automation Fabric, utilizing recipes, data tables, webhooks, and communication connectors:

Component	Workato Implementation	Enterprise Role
Workato Orchestrator	Recipe #82962492 (AegisZero Backend API & Slack Dispatcher)	Serverless event-driven workflow backbone connecting frontend to cloud data.
Workato Data Tables	Table: `Enterprise_Identities` (Imported via CSV)	Persistent database replacing spreadsheets; batch multi-record remediation.
Webhooks Connector	Trigger: Catch Webhook & Action: Return Response	High-speed bidirectional REST gateway accepting Vercel JSON payloads.
Slack Connector	Action: Post message to #security-alerts channel	Branded ChatOps alert card with interactive 4-hr JIT elevation controls.
Email by Workato	Action: Send email to target employee	Automated Zero-Trust offboarding notification with policy rationale.

Multi-Record Batch Remediation Logic in Workato:

When an enterprise triggers remediation, the Vercel frontend compiles a dynamic array of all transferred employees (`drifted_employees`). In Workato, the recipe executes a multi-record update against `Enterprise_Identities`, updating David Chen, Sarah Jenkins, Marcus Vance, and Tariq Al-Mansoor to `REVOKED (Due to Team Change)` and zeroing out admin keys, while leaving Elena Rostova intact as `COMPLIANT`.

4. Quantitative Enterprise Business Impact & ROI

Key Performance Indicator	Before AegisZero	After AegisZero	Enterprise Value Delivered
Audit Velocity	3–4 weeks per quarter	**Sub-second** continuous	120+ manual hours eliminated per quarter.
Zombie SaaS Spend	-$4,625 / mo ($55,500/yr)	**$0.00 waste**	**$55,500.00 / year** direct cash reclaimed.
Toxic Admin Keys	4 unmonitored root keys	**0 standing keys**	100% elimination of root attack vectors.
SOC2 Compliance Posture	18% (Critical Failure)	**98% (Audit-Ready)**	Guaranteed external KPMG/PwC audit clearance.
Pipeline Downtime Risk	High risk of outages	**0.0% Blast Radius**	Deterministic AST safety proof before action.

5. Comprehensive 12-Minute Keynote & Live Demo Script

This verbatim script is calibrated for a complete 12-minute enterprise presentation. Keep this section open on your second screen while operating the live demo on your presentation laptop:

MINUTE 0:00 – 2:00 | Introduction & The Enterprise Problem

[■ Screen: Display https://aegiszero-cockpit.vercel.app. Hover cursor over the 4 red KPI cards and employee table. Do not click buttons.]

"Distinguished judges, enterprise leaders, and the Workato hackathon team: welcome. Today, we are proud to introduce AegisZero—an autonomous multi-agent Zero-Trust identity governance and SaaS FinOps swarm built natively on the Workato Enterprise Automation Fabric.

Every modern technology organization faces an insidious, multi-million-dollar vulnerability that lives in the shadows of their cloud infrastructure: Stale Administrative Privilege Accumulation caused by cross-squad role drift. As engineering organizations grow, engineers frequently transition across squads—moving from payments to data infrastructure, or from platform engineering to customer tooling. While new entitlements are granted instantly, legacy administrative permissions are almost never revoked.

Why does this happen? Because IT administrators operate in sheer terror. They are terrified that if they revoke an AWS root key or a Datadog admin token, they will unknowingly break a background Jenkins CI/CD pipeline, kill an automated database backup daemon, or trigger an outage on production. So they do nothing. Standing administrative keys linger for months.

Look at our live enterprise dashboard right now. Across just five monitored staff members, we have four active Toxic Admin Keys. We are burning $55,500.00 every single year in recoverable zombie spend. And our SOC2 CC6.1 compliance posture is sitting at a disastrous 18%. Take employee David Chen: he transferred to Internal Tooling seventy-five days ago, yet still holds standing AWS Administrator Access. Today, we demonstrate how AegisZero solves this autonomously without human friction."

MINUTE 2:00 – 4:30 | Technical Breakdown of the 8-Agent Swarm

[■ Screen: Scroll to 'Autonomous Swarm War-Room & Deliberations'. Click through Tabs 1, 2, 3, 4, 5, 6, 7, and 8 sequentially.]

"Rather than relying on blunt scripts that cause outages, AegisZero orchestrates an 8-Agent Autonomous Consensus Mesh:

• [Click Tab 1: HR Context Agent]: Ingests HRIS telemetry and discovers that David Chen transitioned squads seventy-five days ago, immediately flagging orphaned approvals.

• [Click Tab 2: Work Entropy Agent]: Analyzes behavioral telemetry. While David logs into SSO daily to maintain appearances, our agent analyzes his Git diffs, database writes, and Figma canvas edits over the last sixty days. The verdict: zero contributions. He has a 94% dormancy score—a passive lurker consuming $1,476 monthly.

• [Click Tab 3: AST Blast-Radius Agent]: This is our crowning mathematical breakthrough. Agent 3 parses the Abstract Syntax Trees of 42 CI/CD pipelines, Dockerfiles, and cron scripts. It searches for hardcoded access tokens and confirms zero dependencies. It proves mathematically that Blast Radius = 0.0%, providing 100% deterministic safety that revoking keys will cause zero downtime.

• [Click Tab 4: FinOps Arbitrage Agent]: Calculates graceful micro-tiering downgrades: AWS Root transitions to zero-cost temporary STS assume-roles; Figma Pro transitions to Free Viewer. Total savings: $17,700/year for David alone.

• [Click Tab 5: SOC2 Compliance Agent]: Formally cites violations of SOC2 CC6.1 and ISO 27001 Annex A.9.2.6.

• [Click Tab 6: JIT & SCIM Synth Agent]: Synthesizes deterministic AWS IAM policy detachment commands and SCIM API patches.

• [Click Tab 7: Slack Bot Comms Agent]: Previews our interactive ChatOps card featuring 1-click self-service 4-hour emergency JIT

elevation.

• [Click Tab 8: Audit Scribe Agent]: Prepares an immutable cryptographic attestation ready for external KPMG and PwC auditors."

MINUTE 4:30 – 7:00 | Live Swarm Execution & Cockpit Transformation

[■ Screen: Scroll up to the top bar. Point cursor to '■ Run 8-Agent Swarm'. Click the button once. Watch gauges turn green.]

"Now, let us watch autonomous enterprise remediation in real time. Watch the metrics closely as I click ■ Run 8-Agent Swarm...

[Click Button: ■ Run 8-Agent Swarm]

In under 800 milliseconds, our cockpit compiles the consensus of all eight agents, constructs a rich JSON payload encompassing all drifted team members, and dispatches it asynchronously into our Workato Workspace.

Witness the live cockpit transformation:
• Our Recoverable Waste immediately updates from a -$4,625/month drain to $55,500.00—100% Reclaimed ROI.
• Our Toxic Admin Keys plummet from 4 down to 0 Keys, with all standing access converted to temporary STS roles.
• Our SOC2 Posture skyrockets from 18% straight to 98% (Audit-Ready)!
• In our directory registry, David Chen, Sarah Jenkins, Marcus Vance, and Tariq Al-Mansoor transition to ● OPTIMIZED, while Elena Rostova remains correctly compliant."

MINUTE 7:00 – 9:30 | Workato Architecture, Data Tables & Slack ChatOps

[■ Screen: Switch tabs to Workato (showing Data Tables & Job History), then to Slack (#security-alerts), then to Email.]

"Let us pull back the curtain and inspect the enterprise infrastructure running live inside Workato:

1. Workato Data Tables: Here in our Enterprise_Identities table, Workato executed an autonomous batch upsert across all transferred team members simultaneously. We have completely eliminated brittle spreadsheets. David Chen, Sarah Jenkins, Marcus Vance, and Tariq Al-Mansoor all reflect status REVOKED (Due to Team Change) with admin keys set to zero.

2. Workato Slack Connector: Here in our Slack #security-alerts channel, our custom bot—AegisZero Security Swarm—posted this comprehensive alert card. Notice the employee-first design: if David Chen faces an urgent production incident, he does not have to file an IT ticket. He can click ■ 1-Click JIT Restore directly inside Slack to receive temporary 4-hour elevated credentials with automatic expiration.

3. Workato Email Connector: Concurrently, Workato dispatched an automated, policy-backed compliance notification to David's inbox detailing the rationale behind his license tier adjustment."

MINUTE 9:30 – 12:00 | Cryptographic SOC2 Audit Export & Conclusion

[■ Screen: Switch back to Vercel. Click '■ Export SOC2 Audit Log'. Open the downloaded JSON certificate file.]

"Finally, let us address the most critical stakeholder in the enterprise: the compliance auditor.

At the top of our cockpit, I click ■ Export SOC2 Audit Log. Instantly, AegisZero downloads a cryptographically signed compliance certificate. Look at the structured JSON: it contains the target employee's role transition history, the 42 CI/CD workflows scanned by our AST engine, the Workato execution timestamp, and an immutable SHA-256 blockchain-ready ledger hash.

To conclude our presentation, AegisZero transforms enterprise identity governance across four fundamental pillars:
1. Time: Replaced 120+ hours of manual quarterly spreadsheet audits with continuous, sub-second orchestration.
2. Money: Reclaimed $55,500.00 every year in unused cloud SaaS seats.
3. Security: Reduced standing root admin credentials from four to zero with mathematical 0.0% downtime proof.
4. Employee Velocity: Replaced days of IT ticket waiting with 1-click self-service Slack JIT elevation.

This is AegisZero: Autonomous Zero-Trust Identity Governance and SaaS FinOps, powered by Workato. Thank you, and we welcome your questions!"